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Predictive insights into U.S. students' mathematics performance on PISA 2022 using ensemble tree-based machine learning models

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ABSTRACT

Background: In the latest Program for International Student Assessment (PISA) 2022 results, U.S. students earned the lowest math scores in two decades. Educators and stakeholders have endeavored to identify key malleable factors in an attempt to raise scores. Although more researchers are gradually incorporating machine learning (ML) techniques, most still rely on literature reviews by humans to identify important predictors. Here we focus on providing innovative insights into how to use ML models to identify predictors most strongly associated with students' math performance.

Methods and Results: The dataset comprises 4,552 U.S. students in 154 schools from the PISA 2022. We used three ensemble tree-based ML models (Random Forest, XGBoost, and LightGBM) to select most influential predictors from 143 derived variables of student and school questionnaires. All three models showed high accuracy in predicting students' math performance, with XGBoost performing best (rMSE = 69.82, training time = 4.14 s) and identifying 10 significant predictors. According to the accumulated local effects (ALEs) plots, three of them have general positive effects, five have roughly negative effects, and two have mixed effects on students' math performance. When comparing these ML-identified predictors to those identified by literature review, the ML method has significantly improved the accuracy of predictor selection (p-value $<$ 0.05) but offered lower interpretability.

Conclusions: We conclude that ML predictor selection is an effective alternative to LR for obtaining influential factors affecting student learning outcomes. Among the factors identified, math self-efficacy, ESCS, and math anxiety are strongly correlate to students' math performance. The results provide valuable insights to implement shifts in instructional practices, targeted

Abbreviations: PISA, Program for International Student Assessment; ML, Machine learning; DVs, Derived variables; IVs, Individual variables; PVs, Plausible values; RF, Random forest; XGBoost, eXtreme gradient boosting; LightGBM, Light gradient-boosting machine.

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interventions, curriculum development, and policy decisions, ultimately contributing to enhancing the overall quality of U.S. math education.

1. Introduction

Despite various reforms and initiatives designed to enhance math education in the United States, challenges persist in achieving consistent student performance. Based on the Program for International Student Assessment (PISA) 2022 results, the U.S. ranked slightly below the Organization for Economic Cooperation and Development (OECD, 2024c) average in mathematics, scoring 465 compared to the OECD average of 472. Although the U.S. score was not drastically different from the average, it consistently lagged behind top-performing countries like Singapore, Hong Kong, and Japan. In the Trends in International Mathematics and Science Study (TIMSS) 2019 results, U.S. 4th graders performed well, with average scoring of 535, which is significantly above the TIMSS average of 500, placing them in the top quarter globally. However, U.S. 8th graders scored 515, still above the average but indicating a relative decline in performance compared to younger students. These results suggest that although the U.S. performs relatively well in early math education, there is a decline in math achievement as students progress to higher grades. In response to these challenges, educators and stakeholders have endeavored to identify key adaptable factors that can overcome them. Scholars have sought to uncover and elucidate the factors contributing to variations in students' mathematics performance by leveraging these vast sources of data. Large-scale assessments play a critical role by providing rich data sources and detailed insights into trends and patterns in students' math performance.

However, most of the papers on international assessment have used traditional statistical methodologies to analyze mathematics performance. In recent years, machine learning (ML) methods have emerged as powerful tools for analyzing educational data and predicting student outcomes with greater precision. ML models excel at uncovering the complex relationships between predictors and responses within large datasets by advanced algorithms. These advantages contribute to more effective decision-making in the field of education. Furthermore, although an increasing number of PISA predictive studies have gradually incorporated ML techniques, most studies selected potential predictors based on literature reviews (LR) conducted by content experts. The LR method may sometimes be influenced by subjective perspectives, potentially overlooking important variables or overestimating those that are less relevant.

The purpose of our study is to fill in the gaps identified in the PISA studies that have remained unexplored. Instead of relying on LR to select influential predictors, we provide innovative insights into how we can use ML models to identify predictors that are strongly associated with students' math performance. The research aims guiding this study are threefold: First, using ML models, we aim to determine which factors are most strongly associated with students' mathematics performance. Second, we seek to explore how ML-derived predictors can be analyzed effectively to provide actionable insights for educators and policymakers. Third, we aim to compare the predictors identified through ML with those reported in the existing PISA literature, highlighting the difference between human selection with theoretical conceptions and selection by ML techniques. The research questions are as follows:

1. What are the most influential predictors associated with students' math performance using ML models?
2. How can we interpret students' math performance using the predictors obtained from the ML model?
3. How do the predictors obtained by ML differ from those in the PISA literature?

To achieve these objectives, we begin by reviewing relevant theoretical framework literature on variables influencing mathematics achievement, including student background characteristics, affective variables, learning opportunities, and school-level variables. We also examine previous studies that have utilized ML approaches to predict students' performance in educational settings. Drawing upon recent frameworks on student and teacher characteristics impacting math achievement, we integrate insights from existing research to inform our analytical framework.

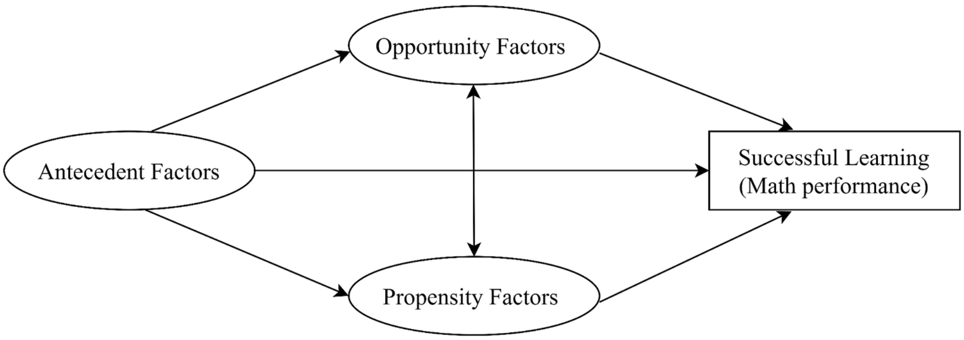

Fig. 1. The opportunity–propensity framework (adapted from Byrnes & Miller, 2007; Powers & An, 2023).

2. Theoretical framework and literature review

2.1. Opportunity–propensity framework

The present study is supported by the opportunity–propensity framework (Byrnes & Miller, 2007; Powers & An, 2023), which is designed to show the relationships among different factors affecting student academic learning, particularly in subjects like mathematics and science (Rubinsten et al., 2018; Wang et al., 2013; Zhu et al., 2018). As shown in Fig. 1, this framework integrates different types of predictors of students' math performance into three categories: antecedent factors, opportunity factors, and propensity factors. Antecedent factors are related to aspects of students' demographic characteristics and family background, such as socioeconomic status and home resources (Desoete & Baten, 2022; Lewis & Farkas, 2017). These factors typically occur earlier in students' mathematical learning and influence their later performance (Liu et al., 2022; Sackes et al., 2011; Star et al., 2015). Further, they are able to affect students' math performance by causing variations in both opportunity factors and propensity factors (Morgan et al., 2023; Ribner et al., 2019).

Opportunity factors refer to aspects of learning context that provide students with opportunities that encourage their mathematical learning, such as school climate and different math-learning activities (Ceulemans et al., 2017; Desoete & Baten, 2022). Prior studies have indicated that students who are in better learning environments with sufficient learning opportunities tend to reach higher academic achievement in mathematics (Powers & An, 2023). Additionally, opportunity factors may impact students' mathematics learning through propensity factors (Baten et al., 2023; Ribner et al., 2019). Propensity factors are students' own willingness and ability to learn mathematics (Hallman, 2014; Mercader et al., 2018), such as students' self-efficacy, motivation, and anxiety. Although antecedent factors and opportunity factors are critical for mathematical learning, students also need the propensity to take advantage of these factors. Propensity factors may influence students' mathematics learning through contributing opportunity factors to students' mathematical learning (Byrnes & Miller-Cotto, 2016; Powers & An, 2023). Finally, these different factors work in an integrated fashion and allow us to understand students' math performance within this framework.

2.2. Factors associated with students' mathematics performance

In the PISA 2022, mathematics performance refers to "the mathematical literacy of a 15-year-old to formulate, employ and interpret mathematics in a variety of contexts to describe, predict and explain phenomena, recognizing the role that mathematics plays in the world." (OECD, 2024a). Four types of factors have been broadly considered to demonstrate an understanding of predicting students' math performance: student background (Lee & Borgonovi, 2022; Marks & Pokropek, 2019), affective factors (Pitsia et al., 2017; Xiao & Sun, 2021), learning opportunities (Guo & Liao, 2022; Wang et al., 2023), and school-related factors, particularly school climate (Ramazan et al., 2023; Teng, 2020). We address these four factors by examining their roles and associations with students' math performance.

2.2.1. Student background

Researchers have frequently explored student background in previous predictive PISA studies on factors affecting math performance. One of the most common factors they have identified as critical in predicting students' math performance is economic, social, and cultural status (ESCS), which is derived from three variables: highest parental occupational status (HISEI), highest education of parents in years (PAREDINT), and home possessions (HOMEPOS) in PISA 2022 (OECD, 2024b). Harju-Luukkainen et al. (2017) examined students' math performance and ESCS background on PISA 2012, and the results indicated that the impact of ESCS on math performance was statistically significant for all students. Based on the PISA 2018 Spanish database, Molina-Muñoz et al. (2023) showed that student ESCS had a significant effect on students' academic performance in mathematics. Not only did these recent studies reveal that student ESCS could significantly predict math performance, some researchers (e.g., Aksu et al., 2022; Masci et al., 2018) found that student ESCS was the most significant factor in predicting math performance in previous PISA cycles.

Similar to the student ESCS, PISA spawned a large body of predictive studies examining the relationship between math performance and another student background factor: information and communication technologies (ICT) resources at home. With the development of technology, specifically artificial intelligence, the role of ICT resources in students' academic success has garnered increasing attention in recent years (Jeong et al., 2024; Lapinid et al., 2022; Zhang & Liu, 2016). Bulut and Cutumisu (2017) and Hu et al. (2018) found that student ICT resources at home were significantly predictive of their math performance on the PISA assessment. Geesa et al. (2019) investigated how students' ICT resources at home predicted their math performance in three countries, including the United States. Their results indicated that ICT availability at home significantly and positively predicted students' math performance. Additionally, some recent studies suggested that the pandemic further increased the importance of students' ICT resources at home in addition to posing enormous educational-equality challenges (OECD, 2023d).

2.2.2. Student affective factors

The affective factors refer to the way in which students relate to their environment, how they perceive it, and how they feel about it (Guo et al., 2020; Jack et al., 2014). The affective factors in PISA consist of subject-specific affective factors and general affective factors (OECD, 2023d). Among math-specific affective factors, multiple studies have demonstrated that math self-efficacy, students' belief in their own ability to succeed in math-related tasks, plays an important role in their math performance (Pitsia et al., 2017; Su et al., 2021). Many predictive PISA studies showed math self-efficacy to be a significant predictor of students' math performance (Arikan, 2014; Koyuncu, 2020; Larsen & Jang, 2021). When Zhao and Ding (2019) examined how factors predicted math performance in the United States and China in PISA 2012, math self-efficacy was the only significant predictor that positively influenced students' math performance across the two countries. By contrast, student math anxiety, characterized by negative emotions and fear responses in solving math tasks, tended to correlate with lower math achievement (D'Agostino et al., 2022; Özberk et al., 2017). As a critical math-specific affective factor, math anxiety showed a significant and negative relationship with students' math performance in previous PISA cycles (Fan et al., 2019; Gabriel et al., 2020; Lau et al., 2022).

Other critical aspects of student affective factors are within the general affective domain, such as student occupational expectations, feelings about learning at home, and out-of-school experiences (Immekus et al., 2022; Michael & Kyriakides, 2023; OECD, 2023c). Many researchers have reported that the factor of occupational expectations was a significant predictor of math performance and that students with high occupation-status expectations usually achieved better math performance than students with low occupation-status expectations (Lee, 2022; Thien, 2016). Furthermore, according to Zhu (2018), the average performance gap in mathematics between students in the top quarter of the PISA occupational status and those in the bottom quarter averages 93 points, which is more than one-and-a-half mathematical proficiency levels.

The affective factor of student feelings about learning at home plays a critical role in students' math performance (Bertling et al., 2020). PISA 2022 results showed nearly half of students stated that they struggled with self-motivation to do schoolwork while learning at home during school closures because of the pandemic (OECD, 2023b), which presented enormous challenges to students' math performance. Suter (2019) found that the factor associated with students' out-of-school experiences had an inverse relationship with students' math performance. Gomez-Fernandez and Albert (2020) also identified a significant negative association between the time spent on exercising or practicing sports before going to school and students' math performance.

2.2.3. *Learning opportunities*

Student learning opportunities—whether students have adequate chances to engage in practical learning experiences—are the most easily neglected factors, yet they play a critical role in predicting students' math performance in PISA assessment (Barnard-Brak et al., 2018; OECD, 2023d; Yang Hansen & Strietholt, 2018). An increasing number of predictive studies have shown a strong positive impact of students' learning opportunities on their mathematics performance, indicating that more learning opportunities lead to better performance in mathematics (Guo & Liao, 2022; Wang et al., 2023). The effects of learning opportunities include the factor of students' familiarity with mathematical concepts, measured by asking students how familiar they are with various mathematical concepts (Costa & Chen, 2023; Özberk et al., 2017; Stacey, 2015). It directly influences students' math performance, with PISA data supporting a positive correlation between familiarity with mathematics and academic results (OECD, 2016a). Some studies even identified familiarity with mathematics concepts as the most significant predictor of students' math performance (e.g., Brow, 2019; Yi & Na, 2020).

Although the literature focuses more on the learning of math concepts in schools, it is equally important to consider the important role of family support on self-directed learning opportunities in students' math performance (Bertling et al., 2020; Liu & Whitford, 2011). Bransford et al. (2000) noted that the average U.S. student spends over half of their academic year learning at home. Moreover, school closures during the pandemic significantly reduced instructional time; therefore, reliance on family support for student self-directed learning, such as a supportive learning environment and help with schoolwork, became more crucial (Grewenig et al., 2021; Manca & Meluzzi, 2020). All of these studies highlighted the importance of family support in promoting self-directed learning, significantly impacting students' math performance.

2.2.4. *School factors*

School-related factors are considered as critical as student-related factors in predicting students' math performance (De Bortoli & Thomson, 2010; Teng, 2020). According to the OECD (2024b) and Wu (2022), the perception of the school principal regarding factors affecting school climate (e.g., student truancy, skipping classes, or bullying other students) could explain a large part of the school climate. In the Bove et al. (2016) PISA 2012 study, the student-related factors affecting school climate showed a significant regression coefficient with math performance at school level. Altun and Kalkan (2021) and Özberk et al. (2017) also determined that the student-related factors affecting school climate were significantly related to students' math performance in previous PISA cycles. Overall, an increasing number of studies acknowledged that a safe and healthy school climate is essential for stimulating students' academic performance (Longobardi et al., 2022; Rohatgi & Scherer, 2020; Trinidad, 2020).

2.3. *Predictive PISA studies on students' math performance using ML*

ML is a subfield of artificial intelligence that concentrates on the development of algorithmic models, enabling computers to automatically learn from large quantities of data to improve predictive accuracy outcomes (Albreiki et al., 2021; Rastrollo-Guerrero et al., 2020; Sekeroglu et al., 2021). Because ML models are able to handle high-dimensional data (King et al., 2024; Martínez-Abad et al., 2020) and learn from the data without relying on predefined, specific distribution assumptions (Cardoso Silva Filho et al., 2023; Liu et al., 2022), there has been a huge increase in the utilization of ML models for predictive analysis of students' math performance in recent years. Gabriel et al. (2018) used the classification and regression decision trees (CART) in PISA 2012 to examine how student background and affective factors relate to math performance. Bayirli et al. (2023) employed different ML models, including random forest (RF), logistic regression (LR), and support vector machine (SVM), to predict the math performance of students from 12 Asia-Pacific countries participating in PISA 2018. Again, in the PISA 2018 Spain database, Gómez-Talal et al. (2024) employed different ML models such as extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), RF, LR, and SVM to identify critical factors predicting students' math performance.

It is worth noting that many of these studies using ML models to predict math performance did not use only a single model; they often used multiple ML models simultaneously. For example, Güre et al. (2020) utilized both Neural Network and RF for predicting Turkish students' math performance in PISA 2015, and the results revealed that RF made predictions with smaller errors. When Tan and Cutumisu (2022) predicted students' academic performance in three subjects, including mathematics, different ML models' predictive results showed that XGBoost slightly outperformed prediction performance. Lee and Lee (2021) identified the important predictors of higher math performance that predict East Asian students' performance; the results showed that LightGBM outperforms the other three ML models. These studies indicate that ensemble tree-based ML models such as RF, XGBoost, and LightGBM have been widely used and exhibited outstanding results in predictive PISA studies on students' math performance.

3. Methods

3.1. Data sources

We analyzed PISA 2022 data (https://www.oecd.org/pisa/data/2022database/), which assessed the mathematical performance of 15-year-old students in the United States. Following the PISA user guidance, we used the school ID (CNTSCHID) to merge the student data file with the school data file. During the merging process, the same school-level predictors were assigned to students within the same school. As a result, the dataset comprised 4552 students in 154 schools. Of those, 2312 were male (50.79 %), 2235 were female (49.10 %), and five students (0.11 %) did not identify their gender. Most of the students were in 10th grade ($N = 3, 378$, 74.21 %), and the rest were in 8th, 9th,11th, and 12th grades ($N = 1174$, 25.79 %).

The student math performance scores were used as outcome variables for the predictive statistical analysis. In PISA 2022, 10 plausible values (PV1MATH to PV10MATH) were provided as math performance scores for each student. To provide unbiased estimates of student proficiency, we followed the user guidance (Kastberg et al., 2021; OECD, 2024c) to include all 10 of these plausible values (PVs) as the outcome variables. In terms of predictors, we used the derived variables (DVs) that were constructed through the arithmetical transformation of one or more multiple items (OECD, 2024b). PISA uses DVs, which provides an advantage in avoiding the issues of collinearity by reducing the overparameterized variability in the values when including multiple predictors measuring the same construct (Rutkowski et al., 2010; von Davier et al., 2019). In PISA 2022, 86 variables were derived from the student questionnaire and 57 from the school questionnaire. This study used all these DVs from the student and school questionnaires as predictors.

3.2. Data analysis

3.2.1. Data cleaning and missing imputation

We performed a series of data preprocessing steps, starting with data cleaning and followed by missing data imputation. In Step 1, we removed predictors with >50 % missing values to prevent biased results due to severe missingness. According to the study by Barton et al. (2023), eliminating predictors with >50 % missing values is essential to prevent biased outcomes when using ML algorithms. After the deletion, the dataset had 70 student-level predictors (originally 86 variables) and 47 school-level predictors (originally 57 variables). In Step 2, we eliminated the identical variables, which had the exact same values for all students in the dataset. Then the dataset had 64 predictors at the student level and 46 predictors at the school level. In Step 3, we eliminated redundant predictors from the PISA 2022 database to avoid the collinearity among variables. For example, ESCS was calculated on the three DVs (HISEI, PAREDINT, HOMEPOS); thus we retained the ESCS and excluded the three DVs representing the ESCS. The dataset then had 48 student-level predictors and 45 school-level predictors. In Step 4, for these remaining variables after data cleaning, we imputed the missing values of the predictors by using the RF-based method—*missForest* imputation. According to Stekhoven and Bühlmann (2012), the RF missing-data imputation method showed approximately 50 % more accuracy than other missing-imputation methods (e.g., regression based, or pattern alternating Lasso) because of the flexibility of multivariate data distribution or variable types. In Step 5, we separated the overall dataset into training (80 % of overall, $N = 3644$) and test (20 % of overall, $N = 908$) datasets.

3.2.2. ML model development

To address the research questions, we used three ensemble tree-based models, including RF, XGBoost, and LightGBM. RF (Breiman, 2001) is a robust and versatile ensemble ML model that works by collecting and evaluating the predictions produced by multiple decision trees using the bagging method (Bayirli et al., 2023; Sokkhey et al., 2020). By aggregating the prediction results obtained from individual trees, the RF reduces prediction variance and improves overall prediction accuracy (Han et al., 2019; Pan & Cutumisu, 2024). XGBoost (Chen & Guestrin, 2016) is another ensemble technique that has gained popularity for its predictive accuracy and computational efficiency (Jeganathan et al., 2022; Tan & Cutumisu, 2022). Unlike RF, XGBoost builds one tree at a time, each tree learning from the differences of the previous one (Asselman et al., 2023; Bentéjac et al., 2021). LightGBM (Ke et al., 2017) employs gradient-based one-side sampling algorithms with the advantages of high accuracy and small memory usage. It integrates the predictions of multiple decision trees to improve the final prediction results (Lee & Lee, 2021; Wang et al., 2022). We conducted the analyses using the *randomForest, xgboost*, and *lightgbm* packages in R software to train three models (Breiman, 2001; Chen & Guestrin, 2016; Ke et al., 2017). Five-fold cross validation was used in the training dataset to improve ML model performance. We conducted our statistical analysis following PISA user guidance (Kastberg et al., 2021; OECD, 2024c). Each analysis was undertaken 10 times and once for each PV, the results were then averaged. For the use of sampling weights, MacNell et al. (2023) pointed out that weighting might be less critical to the ML model configuration process compared to traditional regression models. The findings also demonstrated that ML models still produced reliable results without incorporating sampling weights in a dataset with complex, multistage, and clustered design that use unequal probabilities in the selection process. Building on these, we did not include sampling weights in our analysis.

We obtained the RQ1 results through a series of steps. Following the user guidance, the first step was "*Compute estimates for each plausible value.*" We fitted each of three tree-based models for each of the 10 PVs separately with all predictors. The second step is "*Compute final estimate by averaging all estimates obtained from the first step.*" We computed the final importance estimates of each predictor by averaging importance scores of each predictor. Next, these estimates were ranked in descending order. Then, we started from the most important predictor and employed forward selection to incrementally include additional predictors until satisfying the criterion of absolute relative error (*ARE*; Zhao et al., 2023) that was calculated by

$$ARE = \left| \frac{rMSE_{reduced} - rMSE_{full}}{rMSE_{full}} \right|$$

where rMSE is the root mean squared error between the predicted values and real values, which reflects the predicted accuracy of a model; $rMSE_{full}$ is the rMSE of the model with all predictors; and $rMSE_{reduced}$ is the rMSE of the model including only selected predictors. In this study, we set the *ARE* criterion as 0.1. If *ARE* is less than 0.1, it indicates the reduced model is close to the full model. In other words, the selected predictors are able to explain most of the information from all predictors. For math education, these selected predictors contribute most significantly to improving students' academic performance.

In addition to prediction accuracy, we also reported the execution time of three ML models to compare prediction efficiency because we hypothesized that if the execution time of one model was less than the execution time of the other two models, that model would be more efficient in estimating the parameters of interest. Thus, we identified the best-fitting model based on prediction accuracy and efficiency.

For RQ2, we used the accumulated local effects (ALEs) plots to visualize the relationship between students' math performance and significant predictors selected from the best model in RQ1. The ALE plots (Apley & Zhu, 2020; Tan et al., 2024) show the marginal effects of predictors on the outcome variable based on the conditional distributions when splitting the nodes in ML models. They can help explain how each predictor affects an outcome variable when controlling for all other predictors.

For RQ3, we conducted a literature search based on the key words including "PISA" and "predict" and "mathematics performance" (or "mathematics achievement", "mathematics literacy") from Google Scholar, Education Source, ERIC (EBSCOhost version) and PsycINFO. We included studies whose authors first selected predictors through literature review method and then analyzed the impact of these predictors on students' math performance (e.g., Arikan, 2014; Ding & Zhao, 2019). Regardless of whether their analysis approach involved traditional statistical or ML, if a study's authors employed LR-based method to select predictors, we included the study and counted its predictors as LR-identified predictors. We provided the results of predictor count in the supplementary material. After confirming the LR-identified predictors, we considered the same number of significant predictors as those identified by the best ML model from RQ1, comparing the corresponding rMSE values to reflect the prediction accuracy of LR-identified predictors and ML-identified predictors. Specifically, we first calculated rMSE based on the best ML model with LR-identified predictors or ML-identified predictors for each PV, which measures the root mean squared error between the predicted value and real value; then repeated this calculation for each of the 10 PVs; finally averaged 10 rMSEs to obtain a final rMSE. Additionally, we did a Wilcoxon

Table 1
Selected predictors based on three ML models.

ML Models	Selected Predictors	ARE (%)
RF	MATHEFF (Mathematics self-efficacy)	9.62
	ESCS (Index of economic, social and cultural status)	
	FAMCON (Familiarity with mathematics concepts)	
	BSMJ (Expected occupation status)	
	FAMSUPSL (Family support for self-directed learning)	
XGBoost	MATHEFF (Mathematics self-efficacy)	9.79
	ESCS (Index of economic, social and cultural status)	
	FAMCON (Familiarity with mathematics concepts)	
	BSMJ (Expected occupation status)	
	FAMSUPSL (Family support for self-directed learning)	
	WORKPAY (Working for pay before or after school)	
	FEELLAH (Feelings about learning at home)	
	EXERPRAC (Exercising or practicing a sport before or after school)	
	ANXMAT (Mathematics anxiety)	
	ICTRES (ICT resources at home)	
LightGBM	MATHEFF (Mathematics self-efficacy)	9.81
	ESCS (Index of economic, social and cultural status)	
	BSMJ (Expected occupation status)	
	FAMCON (Familiarity with mathematics concepts)	
	FAMSUPSL (Family support for self-directed learning)	
	WORKPAY (Working for pay before or after school)	
	FEELLAH (Feelings about learning at home)	
	EXERPRAC (Exercising or practicing a sport before or after school)	
	ANXMAT (Mathematics anxiety)	
	ICTRES (ICT resources at home)	

signed-rank test (Woolson, 2007) to examine whether there was a statistical difference between the two predictor selection methods. We also discussed the efficiency and interpretability to analyze the advantages and weaknesses of each method in this study.

4. Results

4.1. RQ1. What are the most influential predictors associated with students' math performance using ML models?

We examined the significant predictors of students' math performance in three ML models. Each ML model selected a different number of predictors: RF selected five predictors with an *ARE* value of 9.62 %, XGBoost selected ten predictors with an *ARE* value of 9.79 %, and LightGBM selected ten predictors with an *ARE* value of 9.81 % (see Table 1). These predictors are listed by relative ranking, which indicates how more or less important each predictor was in influencing students' math performance. As the first two significant predictors, MATHEFF refers to student self-efficacy to perform various math-related tasks, and ESCS is a complex composite index that measures students' economic, social, and cultural status. Three predictors (FAMCON, BSMJ, and FAMSUPSL) rank among the top three to five predictors. FAMCON measures student familiarity with mathematical concepts like exponential function and three-dimensional geometry. BSMJ refers to students' expected occupation status, that is, what kind of job they expect to have when they are approximately 30 years old. FAMSUPSL refers to how often students' families provide specific kinds of learning support such as creating a learning schedule. These top five were selected as significant predictors in all three ML models.

The other five (WORKPAY, FEELLAH, ICTRES, ANXMAT, EXERPRAC) were selected as significant predictors in XGBoost and LightGBM, and all of them follow the same ranking sequences in both ML models. WORKPAY refers to how many days during a typical school week students work for pay before going to school or after leaving school. FEELLAH refers to how students felt about learning at home, for example, "I enjoyed learning by myself." EXERPRAC indicates how many days during a typical school week students exercised or practiced a sport before or after school, similar to WORKPAY. ANXMAT indicates how anxious students are toward math, for example. "I often worry that it will be difficult for me in mathematics classes." ICTRES refers to students' access to ICT at home, for example., "I own a cell phone with internet access in my home."

In Table 2, we provided prediction accuracy (rMSE) and efficiency (training time) as metrics to compare the performance of three ML models in selecting predictors strongly associated with math performance. The full model includes all predictors. The reduced model includes only selected predictors — 5 predictors in RF, 10 predictors in XGBoost, and 10 predictors in LightGBM. Based on the small differences of rMSE between the full model and reduced model under all three ML models, we identified that all three ML models have high accuracy when using the selected predictors instead of all predictors to predict students' math performance. Comparing the rMSE values for each reduced model, XGBoost (rMSE = 66.39) and LightGBM (rMSE = 66.36) had less prediction error than RF (rMSE = 69.82). In terms of efficiency in data analysis, XGBoost (training time = 4.14 s) demonstrated shorter execution times than LightGBM (training time = 22.93 s.) and RF (training time = 129.27 s.). Thus, by considering both accuracy and efficiency, we identified XGBoost as the best model for predicting PISA 2022 U.S. students' math performance in this study.

4.2. RQ2. How can we interpret students' math performance using the predictors obtained from the ML model?

To balance the interpretability of ML techniques, we used the ALEs plot to visualize the relationship between students' math performance and significant predictors selected by XGBoost. As shown in Fig. 2, math self-efficacy (MATHEFF) shows a strong positive impact on math performance; students with higher math self-efficacy tend to perform better in math tasks. This indicates the importance of students' confidence in their math abilities. A student's ESCS background also has a positive effect on math performance, which aligns with prior findings that higher socioeconomic backgrounds could provide better resources for students' math learning success. Importantly, the math performance plateaus at the maximum point around a value of 2 and becomes flat, which means increasingly smaller benefits of socioeconomic in math learning after achieving a certain level. Similarly, familiarity with mathematics concepts (FAMCON) is positively correlated with math performance, particularly within the range of −2.5 to 2.5. This demonstrates that a deeper understanding of math concepts supports better math learning outcomes, especially for most foundational math tasks. These significant predictors emphasize the three critical areas where interventions can improve students' math performance by increasing self-efficacy, socioeconomic background, and understanding of mathematical concepts.

There are five predictors (FAMSUPSL, WORKDAY, FEELLAH, ANXMAT, and EXERPRAC) that demonstrate roughly negative effects on students' math performance. Specifically, both family support for self-directed learning (FAMSUPSL) and feelings about learning at

Table 2
Model comparison based on prediction accuracy and efficiency.

ML Models	Metrics	Full Model	Reduced Model	Number of Selected Predictors
RF	rMSE	63.69	69.82	5
	Training Time	1098.72	129.27	
XGBoost	rMSE	60.47	66.39	10
	Training Time	8.23	4.14	
LightGBM	rMSE	60.43	66.36	10
	Training Time	44.28	22.93	

Note. rMSE = root mean squared error.

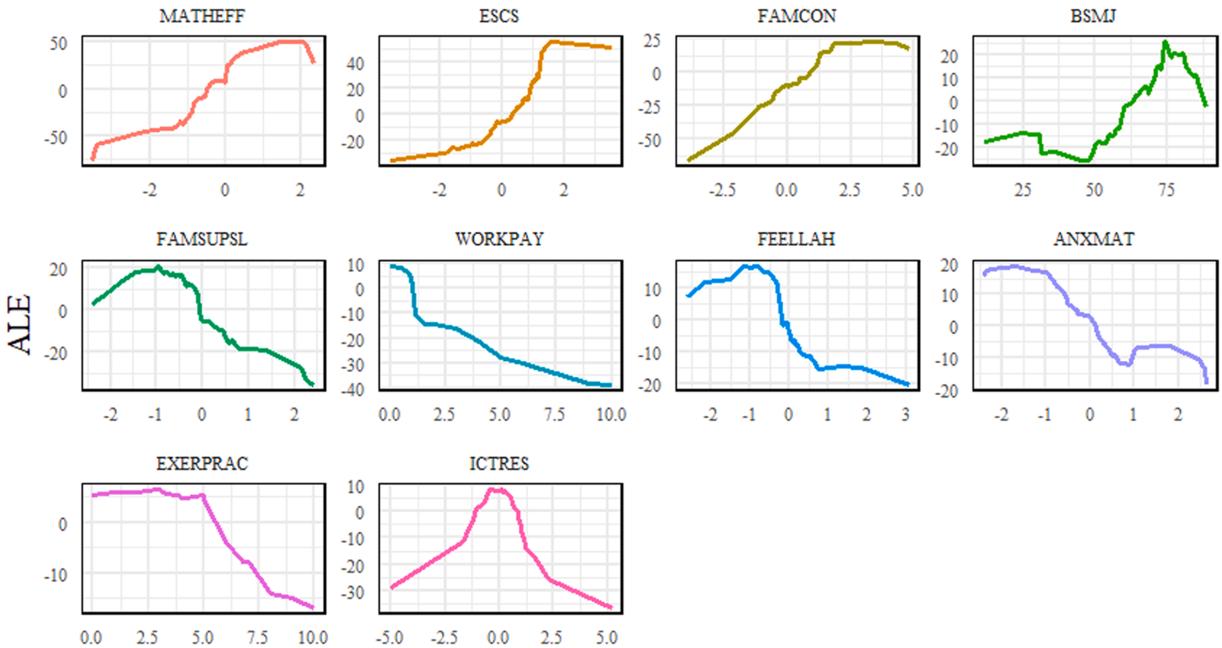

Fig. 2. ALE Plots of Selected Predictors from the XGBoost Model. Note. The y-axis represents the ALE values of students' math performance; the x-axis represents the value of the predictors.

home (FEELLAH) show a slight increase in math performance when they are at small values. These indicate that a small amount of family support or positive feelings about learning at home can improve math performance. However, the short-term benefits were quickly followed by overall declining overall trends. For family support, this suggests that although a small amount of family involvement can be beneficial for math learning, too much support might hinder a student's ability to learn independently. Regarding feelings about learning at home, the decline indicates that although a positive home learning environment can initially improve achievement, excessive attention or pressure on home learning may ultimately be counterproductive. In terms of ANXMAT, there is a slight improvement in math performance that occurs at approximately one-third of the measurement scale. This indicates that appropriate anxiety may be helpful for students' math performance, but higher anxiety has a generally negative effect. The ALE plots for working for pay before or after school (WORKPAY) and exercising or practicing a sport before or after school (EXERPRAC) also show negative effects on math performance. Students who spend more time working before or after school are likely to struggle in math learning. This might be because they have less time and energy available for academic tasks. Similarly, excessive exercising or sports practices before or after school might take attention away from academic learning and result in lower math achievement.

Additionally, BSMJ represents a skewed S-shaped pattern with an overall upward trend; however, there are declining trends when BSMJ is at approximately the one-fourth and three-fourths positions of the measurement scale. This suggests that too high or too low an expected occupation status negatively affects students' math performance. Finally, ICTRES is a symmetrical, hat-shaped pattern in which math performance goes up with negative domain and down with positive domain. This demonstrates that moderate ICT resources at home can be beneficial but overuse will have negative effects on students' math performance.

Table 3
Top 10 predictors based on literature review.

Rank	Selected Predictors	References
1	MATHEFF (Mathematics self-efficacy)	#2, #3, #6, #7, #8, #10, #12, #13, #15, #17, #19, #21, #22, #23, #25, #26, #28
2	Gender	#1, #2, #3, #4, #5, #9, #11, #12, #14, #15, #18, #19, #20, #22
3	ANXMAT (Mathematics anxiety)	#2, #3, #7, #8, #13, #15, #17, #19, #22; #23, #25, #26, #28
4	ESCS (Index of economic, social and cultural status)	#1, #2, #3, #5, #9, #10, #14, #16, #18, #19, #22
5	IMMIG (Immigration background)	#3, #11, #15, #16, #18, #20, #21
6	MATHMOT (Motivation)	#2, #19, #23, #24, #28, #29
7	TEACHSUP (Teacher support)	#7, #8, #17, #21, #22, #23
8	STRATIO (Student–teacher ratio)	#9, #11, #16, #17, #18, #21
9	REPEAT (Grade repetition)	#15, #16, #17, #18, #20
10	BELONG (Sense of belonging)	#1, #8, #21, #23, #26

Note: The reference lists can be found in the supplementary material.

4.3. RQ3. How do the predictors obtained by ML differ from those in the PISA literature?

To answer RQ3, we first conducted a literature search where we found a total of 29 studies focused on predictive analysis to explore various factors that affect students' math performance. We found 67 important predictors from PISA 2003 to PISA 2018 in the 29 studies, as shown in the supplementary material. Because the 10th predictor (mathematics interest) is not able to be matched to the same or similar variables in PISA 2022, we skipped it and selected the 11th predictor (sense of belonging) as the new 10th predictor. Finally, we listed the top 10 predictors in Table 3. There are three predictors (MATHEFF, ANXMAT, and ESCS) commonly selected by ML and the LR

Fig. 3 shows the accuracy trends of predictors selected by ML and those selected by LR. The two predictor selection methods start at the same rMSE value because MATHEFF is the most important predictor selected by both of them. As the number of predictors increases, both predictor selection methods show a decreasing trend in rMSE, indicating that the prediction accuracy consistently improves as more predictors are included. Specifically, we can see the predictors from ML and LR result in different prediction accuracy under the same numbers with different types. For example, the top two predictors selected by ML (MATHEFF and ESCS) have an rMSE of 78.75, and the top two selected by LR (MATHEFF and Gender) have an rMSE of 83.28. Therefore, the predictors from ML produce more accurate results than the predictors from LR. Further, we conducted a Wilcoxon signed-rank test to examine whether there was a statistical difference between the two predictor selection methods. The test results showed a *p*-value of <0.05. From accuracy perspective, this indicated that the ML method significantly improved predictor selection in the PISA 2022 U.S. dataset.

5. Discussion

5.1. Implications of the results

This study provides innovative insights into using ML predictor selection to obtain the factors most strongly associated with students' mathematics performance. All three ensemble tree-based ML models show high accuracy when using selected predictors instead of all predictors to predict math performance. Moreover, there is a significant overlap of predictors under the three ML models, which demonstrates the consistency of ML predictor selection results.

The study reveals that the ML method provides a significant improvement in predictor selection over the traditional literature search of previous studies, especially in large-scale assessment contexts. Compared to literature searches, which incur high costs in time and human expertise, ML has contributed to an improvement in efficiency. However, ML processes are like a "black box" because of a lack of interpretability (Liu et al., 2022; Puah, 2020). The LR method usually obtains potential important factors based on experts' knowledge or established-theory frameworks, which provide high-quality interpretability of predictors. As a data-driven method, ML may find complex patterns in the data that may not immediately be found through LR. Aligned with the study of Immekus et al. (2022), the application of ML techniques to large-scale assessment data can unlock much potential to advance theory and research. By considering accuracy, efficiency, and interpretability, both methods are valuable for helping to identify important key predictors. ML is an effective alternative to LR for obtaining the influential factors that impact student math learning outcomes.

In this study, the significant predictors identified through ML can be categorized into three types of factors—antecedent, opportunity, and propensity—according to the opportunity–propensity framework. For antecedent factors, student ESCS and ICT resources

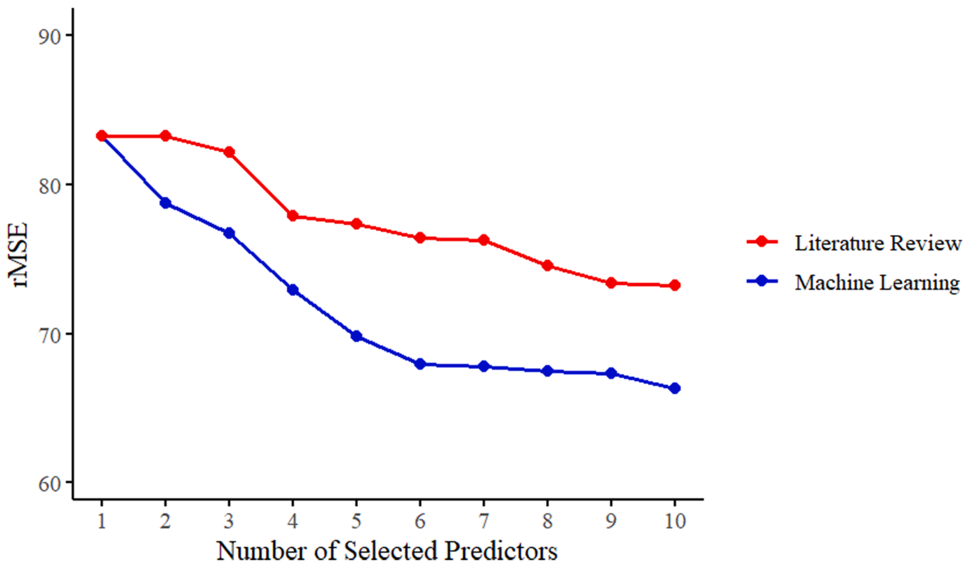

Fig. 3. Comparison of prediction accuracy between predictors identified through literature review and machine learning model. *Note.* rMSE = root mean squared error.

at home stand out as two influential factors in predicting students' math performance. As the second ranking predictor, ESCS had a significant and positive effect on students' academic performance in math. Its most important role and positive impact on student mathematical learning outcomes is similar to science and reading disciplines in PISA (Vazquez-Lopez & Huerta-Manzanilla, 2021; Yetişir & Batı, 2021). Unlike ESCS, the effect of ICT resources at home on student math learning is both positive and negative. Using these resources (e.g., laptop, tablet, and smartphone) appropriately will help improve students' math performance; however, excessive use will have a negative impact. In the findings of Gómez-García et al. (2020), a total of 64 % of secondary students affirmed that they used ICT at home to study mathematics, but the availability of ICT resources is often negatively associated with their success because of inappropriate or excessive use such as frequent phone calls and social media. Some researchers also revealed that ICT resources would influence students' mathematical learning through affecting their perceptions (Courtney et al., 2022; Fan et al., 2022). It is worth highlighting that the dual impact of ICT resources emphasizes the need for balance to maximize its positive effects on students' math learning.

Within opportunity factors, the positive impact of "familiarity with mathematical concepts" highlights that students who are exposed to more mathematical content in their mathematics lessons tend to perform better. Weaker relative performance in a content area might signal an imbalance in the mathematical curriculum, which could result in curriculum reform (OECD, 2016b). In addition, greater familiarity with mathematics may not be sufficient for solving the most complex mathematics tasks. It is essential for students to be exposed to problems that stimulate their reasoning abilities and promote their conceptual understanding, creativity, and problem-solving skills. Additionally, the global spread of the pandemic in 2020 disrupted normal teaching and learning activities (OECD, 2023d). During school closures, students' math performances may suffer because too much family support might hinder their independent ability for self-directed learning at home. Without proper family involvement, students may face difficulties in staying motivated, completing assignments, and understanding mathematical concepts (Toala-Bailón et al., 2022).

For propensity factors, student self-efficacy and anxiety are two influential predictors that significantly affect academic performance in mathematics domains. Mathematics self-efficacy played the most positive predictive role in affecting students' math performance. Conversely, high math anxiety generally showed a negative trend, with higher anxiety typically linked to lower math performance. That said, it is interesting that appropriate anxiety has potential benefits in improving math performance. The study by Luttenberger et al. (2018) revealed that appropriate anxiety can contribute to mathematical learning by providing a slight level of motivation and focus. In the general domain, the factor of student occupational expectations ranks as the fourth most significant predictor positively affecting math performance. This suggests that students with high future career expectations usually perform better in math. Moreover, student engagement in school or out-of-school experiences (e.g., exercising and working for pay) is another significant factor in their math achievement. Not only for school, the factor of "student feeling about learning at home" is also a significant factor in their ability to learn math. Especially because of the pandemic, many students struggled to have positive feelings about learning at home during school closures, which significantly and negatively affected their math performance.

5.2. Practical approaches to instruction to improve math performance

According to the latest data from the PISA 2022, U.S. students performed above the OECD average in science and reading but lagged behind in math. The results revealed a statistically significant decline in U.S. students' math performance since 2018, marking the lowest PISA math scores for the country in the past two decades (OECD, 2023c). This downward trend in math performance underscores the urgency for educational stakeholders to identify and address the key factors influencing student outcomes. Numerous initiatives have been undertaken to enhance math learning among U.S. students, yet ongoing challenges highlight the need for more targeted and effective approaches. In this study we examined critical factors influencing math performance. The most significant factor among the student-related variables was self-efficacy. Self-efficacy plays a crucial role in influencing students' motivation, engagement, and academic success. A substantial body of research has consistently demonstrated the positive correlation between self-efficacy in mathematical tasks and overall math performance (Arens et al., 2022). Students with higher self-efficacy in their math abilities tend to perform better on assignments and tests than those with lower self-efficacy (Özcan & Kültür, 2021). In fact, numerous studies have shown that self-efficacy is a strong predictor of math performance, even after accounting for other factors such as SES and grades (e.g., Hiller et al. 2022).

Prior knowledge of math concepts could be the essential foundation upon which all new mathematical understanding is constructed (Clements & Sarama, 2020). When students develop a robust base in fundamental math principles, they are distinctly better equipped to integrate and apply new concepts and knowledge effectively, they accelerate their learning process, and they significantly improve their overall academic achievement. Conversely, a lack of adequate prior knowledge can create a barrier to moving forward, resulting in frustration and disengagement from math. To address this issue, it is critical to assess and subsequently target gaps in students' foundational math knowledge. Early identification of these gaps allows for timely interventions that prevent future learning difficulties.

This study also highlights the significant impact of career expectations on math performance. Teachers and educators play a crucial role in bridging the gap between mathematical skills and their application in future careers, particularly in fields that heavily rely on math, such as engineering, finance, technology, and science. By integrating real-world career scenarios into math lessons, teachers can enhance the relevance and appeal of the subject, thereby boosting students' interest and engagement. Such educational strategies not only deepen students' understanding of math but also allow them to develop essential skills like problem-solving, teamwork, and project management. Through these methods, teachers and educators can significantly increase students' motivation related to their career aspirations, transforming math education into a more dynamic and goal-oriented experience that aligns with their future professional goals.

5.3. Methodological implication

Our results showed high accuracy and efficacy in predicting students' math performance, supporting the applicability of the ML models. In previous research, predictor selection in large-scale assessments relied heavily on literature searches. By using ML techniques, researchers are able to leverage the data-driven method to automatically identify the most influential predictors related to students' academic performance. As a data-driven method, ML is good at processing large amounts of data very quickly and identifying patterns that may not immediately be found by human experts (Kumar & Sharma, 2022; Sturm et al., 2021). Furthermore, it is critical to incorporate ML techniques with interpretability tools, such as ALE plots, to ensure the results are grounded in educational contexts. In the future, complementing human expertise with ML techniques will provide more comprehensive understanding of student performance and support more effective educational interventions.

According to Iatrellis et al. (2021), ML methods are based on data-driven approaches, which depend on statistical criteria such as rMSE. The ML methods derive the model parameters by iterating the multiple times of estimation process (either sequential steps for the XGboost model or parallel steps for the RF model). The iterative model estimation improves prediction accuracy as well. Thus, future researchers who will potentially build a prediction model for educational outcomes could consider using the ML methods rather than traditional prediction models such as linear or logistic regression models (Asselman et al., 2023; Jeganathan et al., 2022).

In addition, we demonstrated that the ML models can be used as a useful tool for the missing data imputation method applying RF framework. As suggested by Stekhoven and Bühlmann (2012), because the RF-based missing data imputation is nonparametric, it performs better when the data violate the parametric missing data imputation, especially for nonnormal data. From the PISA 2022 report, the missing data rates for overall variables were approximately 70 %. In the case of severely skewed data distribution, the regression-based missing data imputation produces a biased imputed value, which causes errors in statistical analysis results. Hence, when using PISA data with any violation of the parametric missing data technique, we could consider using ML-based methods such as RF missing data imputation as an alternative.

5.4. Limitations and future study

In this study, the predictors identified by the LR method were derived from an analysis of 29 studies. Despite our rigorous search strategy, which involved the use of specific keywords and a range of databases detailed in the methods section, there remains a possibility that we may have inadvertently excluded some relevant papers. This risk of missing eligible studies is an inherent limitation of any LR process, and the potential omission of important studies could affect finding the identified predictors and may lead to a less robust understanding of the factors influencing the outcomes under investigation. To mitigate this risk, future researchers could employ additional strategies such as citation tracking to ensure a more exhaustive collection of pertinent literature.

To enhance the generalizability of the results, future researchers could use ML models on cross-national datasets to systematically compare predictors of math performance across various countries. By analyzing these diverse datasets, we could identify both universal factors that consistently influence math achievement and unique, country-specific factors that vary by region. This methodological approach would not only reveal underlying patterns in global math education but also provide insights into specific national educational practices. Such findings could support understanding the global trends and regions' needs in math education.

6. Conclusion

Large-scale assessments provide valuable data and insights into student performance and the factors that impact learning outcomes. A prediction model derived from the data shows trends and disparities in performance related to different factors. With advanced ML techniques, our goal is to provide practical, data-driven recommendations for improving educational outcomes. Specifically, our research is focused on identifying the most influential predictors of math performance and comparing these ML-identified predictors to those traditionally identified. The results provide valuable insights for shifts of instructional practices, targeted interventions, curriculum development, and policy decisions. This would ultimately contribute to the overall quality of math education in the United States.

CRediT authorship contribution statement

Li Zhu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Hyesun You:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Conceptualization. **Minju Hong:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Zhenhan Fang:** Writing – review & editing, Writing – original draft, Software, Methodology.

Declaration of competing interest

The authors declare that they have no competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijer.2025.102537.

Data availability

The datasets used in the current study are available upon reasonable request.

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